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**BCA (Sem-V) – Micro-
processor (503) DSE-I**

2022

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section–A

Objective type Questions

1. Choose the correct alternative for each of the
following questions : **2×10=20**

(a) Which of the following technologies was
used by Intel to design its first 8-bit

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(1)

(Turn over)

microprocessor?

- ☒ (i) NMOS
 - (ii) HMOS
 - ☒ (iii) PMOS
 - (iv) TTL
- (b) Which of the following is not true about the address bus ?
- (i) It is 32 bits in length
 - ☒ (ii) It is a bidirectional bus
 - (iii) It is 16 bits in length
 - (iv) Lower address bus lines ($AD_0 - AD_7$) are called "Line number"
- (c) Which of the following interrupts has the highest priority ?
- ☒ (i) TRAP
 - (ii) INTR

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(2)

Contd.

(iii) RST6.5

(iv) None of these

(d) Which of the following flag is used to mask INTR interrupt ?

- (i) Zero flag
- (ii) Auxiliary carry flag
- ☒ (iii) Interrupt flag
- (iv) Sign flag

(e) Which of the following is a special-purpose register for a microprocessor ?

- (i) Program counter
- ☒ (ii) Instruction register
- (iii) Accumulator
- (iv) Temporary register

(f) How many flip-flops are there in a flag register of an 8085 microprocessor ?

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(3)

(Turn over)

- (i) 4
 - ~~(ii)~~ 5
 - (iii) 7
 - (iv) 10
- (g) Which of the following is not a status flag in microprocessor ?
- (i) Zero flag
 - (ii) Sign flag
 - (iii) Parity flag
 - ~~(iv)~~ Index flag
- (h) What is stored in the HL pair register ?
- (i) Opcode
 - ~~(ii)~~ Address of memory
 - (iii) Address of next instruction
 - ~~(iv)~~ Temporary data
- (i) Which of the following is true about stack pointers ?

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(4)

Contd.

- ~~(i)~~ Stack pointer contains the address of the top of the stack memory
 - (ii) Stack pointer is an 8-bit register
 - (iii) Stack pointer stores data permanently
 - (iv) Stack pointer is initialized after stack operation
- (j) Which of the following is true about MOV A, B instruction ?
- (i) It means move the content of register A to register B
 - ~~(ii)~~ It means move the content of register B to register A
 - (iii) It doesn't affect the flag register
 - (iv) None of these

Section-B

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(5)

(Turn over)

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Write the different types of interrupts.
3. What is the immediate addressing mode ?
Give an example.
4. Write assembly language instructions for the following:
 - (a) Move the content of memory to register.
 - (b) Move immediate data to memory.
5. What are general purpose registers?
6. Write the bus structure of the microprocessor.
7. What is a video controller ?

Section-C

Long Answer type Questions

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(6)

Contd.

Answer any **two** questions of the following :

15×2=30

8. Explain 8257 DMA controller.
11. Write an assembly language program to Add 49H and 56H.
The first number 49H is in the memory location 2501H.
The first number 56H is in the memory location 2502H.
The result is to be stored in the memory location 2503H.
12. Explain different types of Status Flag registers.
13. Explain memory address decoding.

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(7)

(P-300)